

Stochastic Coordinate Descent for Polyak-Lojasiewicz Functions

1 Algorithm Description

The algorithm iteratively updates the full parameter vector by computing the gradient along a randomly selected coordinate direction at each step. It uses this coordinate-wise gradient information to determine the update direction and step size for the entire parameter vector. It is designed for functions satisfying the Polyak-Lojasiewicz condition that are smooth with respect to individual coordinates.

1.1 Mathematical Formulation

The algorithm is a variant of stochastic coordinate descent, designed for functions that satisfy the Polyak-Lojasiewicz (PL) condition and are smooth with respect to individual coordinates. The PL condition is a relaxation of strong convexity, providing a sufficient condition for convergence.

Consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies:

1. Polyak-Lojasiewicz Condition:

$$\frac{1}{2} \|\nabla f(w)\|^2 \geq \mu(f(w) - f^*)$$

where $\mu > 0$ and f^* is the minimum value of f .

2. Coordinate-wise Smoothness: For each coordinate i , there exists $L_i > 0$ such that:

$$\left| \frac{\partial f(w)}{\partial w_i} - \frac{\partial f(v)}{\partial w_i} \right| \leq L_i |w_i - v_i|$$

1.2 Algorithm Steps

- **Initialization:** Start with an initial parameter vector w_0 .

- **Iteration:**

1. Randomly select a coordinate $i \in \{1, 2, \dots, n\}$.
2. Compute the partial gradient: $g_i = \frac{\partial f(w)}{\partial w_i}$.
3. Update the parameter vector:

$$w_i \leftarrow w_i - \eta g_i$$

4. η is the step size, a hyperparameter controlling the convergence rate.

- **Termination:** Continue until convergence criteria are met, e.g., $\|\nabla f(w)\| < \epsilon$ for some small $\epsilon > 0$.

2 Pseudo Code and Dry Run

2.1 Pseudocode

```
Initialize  $w = w_0$ 
Set step size  $\eta$ 
Set convergence tolerance  $\varepsilon$ 

Repeat until convergence:
    Randomly select coordinate  $i$  from  $\{1, 2, \dots, n\}$ 
    Compute gradient  $g_i = \partial f(w) / \partial w_i$ 
    Update  $w_i = w_i - \eta * g_i$ 
    Check if  $|g_i| < \varepsilon$  for all  $i$ 

Return  $w$ 
```

2.2 Dry Run Example

Consider the quadratic function $f(w) = (w - 3)^2$.

Initialization:

- Start with $w_0 = 0$.
- Set step size $\eta = 0.1$.

Iterations:

1. Iteration 1:

- Gradient: $g = \frac{d}{dw}(w - 3)^2 = 2(w - 3)$.
- At $w_0 = 0$, $g = 2(0 - 3) = -6$.
- Update $w_1 = w_0 - \eta g = 0 - 0.1 \times (-6) = 0.6$.
- Objective value: $f(w_1) = (0.6 - 3)^2 = 5.76$.

2. Iteration 2:

- Gradient: $g = 2(0.6 - 3) = -4.8$.
- Update $w_2 = w_1 - \eta g = 0.6 - 0.1 \times (-4.8) = 1.08$.
- Objective value: $f(w_2) = (1.08 - 3)^2 = 3.6864$.

3. Iteration 3:

- Gradient: $g = 2(1.08 - 3) = -3.84$.
- Update $w_3 = w_2 - \eta g = 1.08 - 0.1 \times (-3.84) = 1.464$.
- Objective value: $f(w_3) = (1.464 - 3)^2 = 2.3593$.

The parameter vector w is progressively moving towards the optimal value of 3, with the objective value decreasing at each step.

3 Convergence Theory

3.1 Assumptions

We consider the following assumptions for our algorithm:

1. **Polyak-Lojasiewicz (PL) Condition:** The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfies the PL condition if there exists a constant $\mu > 0$ such that for all $\mathbf{x} \in \mathbb{R}^n$,

$$\frac{1}{2} \|\nabla f(\mathbf{x})\|^2 \geq \mu(f(\mathbf{x}) - f^*),$$

where f^* is the optimal function value.

2. **Coordinate-wise Smoothness:** The function f is coordinate-wise L -smooth if for each coordinate i , there exists a constant $L_i > 0$ such that for all $\mathbf{x} \in \mathbb{R}^n$,

$$\left| \frac{\partial f(\mathbf{x} + \alpha \mathbf{e}_i)}{\partial x_i} - \frac{\partial f(\mathbf{x})}{\partial x_i} \right| \leq L_i |\alpha|,$$

where $\mathbf{e}_i$ is the unit vector in the i -th coordinate direction.

3. **Bounded Gradient:** The gradient is assumed to be bounded, i.e., there exists $G > 0$ such that $\|\nabla f(\mathbf{x})\| \leq G$ for all $\mathbf{x} \in \mathbb{R}^n$.
4. **Random Coordinate Selection:** At each iteration, a coordinate i is selected uniformly at random from $\{1, 2, \dots, n\}$.

3.2 Main Convergence Theorem

Under the above assumptions, the algorithm converges with the following properties:

For a function f satisfying the PL condition and coordinate-wise smoothness, consider the iterative update

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta_k \nabla_i f(\mathbf{x}_k) \mathbf{e}_i,$$

where i is chosen uniformly at random, η_k is a suitable step size. Then, the algorithm achieves a convergence rate of $O(1/T)$ in expectation, given by

$$\mathbb{E}[f(\mathbf{x}_T)] - f^* \leq \frac{C}{T},$$

for some constant $C > 0$, where T is the number of iterations.

3.3 Theoretical Properties

1. **Stability Analysis:** The algorithm is stable under the choice of step size $\eta_k \leq \frac{1}{L}$, where $L = \max_i L_i$. This ensures that the updates do not diverge.
2. **Sensitivity to Hyperparameters:** The convergence rate is sensitive to the choice of step size η_k . A smaller step size ensures stability, whereas a larger step size can speed up convergence at the risk of instability.
3. **Comparison to Baseline Methods:** Compared to standard gradient descent, this coordinate-wise approach can be more efficient in high-dimensional spaces, as it reduces the computational cost per iteration by focusing on one coordinate at a time.

3.4 Discussion

In this document, we have developed a convergence theory for a coordinate-wise gradient descent algorithm under the PL condition and coordinate-wise smoothness. The main result shows that the algorithm achieves a sublinear rate of convergence, consistent with known results for similar methods under these conditions. The theoretical properties highlight the algorithm's stability and sensitivity to hyperparameters, suggesting that careful selection of step size is crucial for optimal performance. The analysis also shows that the algorithm can be particularly beneficial in scenarios involving high-dimensional functions, offering computational efficiency over traditional full-gradient methods.

4 Convergence Proof

4.1 Preliminaries (Definitions and Lemmas)

4.1.1 Definitions

We consider a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ with the following properties:

1. **Polyak-Lojasiewicz (PL) Condition:** There exists $\mu > 0$ such that for all $\mathbf{x} \in \mathbb{R}^n$,

$$\frac{1}{2} \|\nabla f(\mathbf{x})\|^2 \geq \mu(f(\mathbf{x}) - f^*),$$

where f^* is the optimal function value.

2. **Coordinate-wise Smoothness:** For each coordinate i , there exists $L_i > 0$ such that for all $\mathbf{x} \in \mathbb{R}^n$,

$$\left| \frac{\partial f(\mathbf{x} + \alpha \mathbf{e}_i)}{\partial x_i} - \frac{\partial f(\mathbf{x})}{\partial x_i} \right| \leq L_i |\alpha|,$$

where $\mathbf{e}_i$ is the unit vector in the i -th coordinate direction.

3. **Bounded Gradient:** There exists $G > 0$ such that $\|\nabla f(\mathbf{x})\| \leq G$ for all $\mathbf{x} \in \mathbb{R}^n$.

4. **Random Coordinate Selection:** At each iteration, a coordinate i is selected uniformly at random from $\{1, 2, \dots, n\}$.

4.1.2 Lemmas

For a function f satisfying the PL condition, the following holds for any $\mathbf{x} \in \mathbb{R}^n$:

$$f(\mathbf{x}) - f^* \leq \frac{1}{2\mu} \|\nabla f(\mathbf{x})\|^2.$$

For a coordinate-wise L -smooth function, if i is chosen uniformly at random, the expected squared norm of the gradient is bounded by:

$$\mathbb{E}[\|\nabla_i f(\mathbf{x})\|^2] \leq \frac{1}{n} \|\nabla f(\mathbf{x})\|^2.$$

4.2 Main Proof (Step-by-Step Derivation)

Consider the iterative update:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \eta_k \nabla_i f(\mathbf{x}_k) \mathbf{e}_i,$$

where i is chosen uniformly at random.

4.2.1 Step 1: Expansion of the Update Rule

The update rule can be expanded as:

$$f(\mathbf{x}_{k+1}) = f(\mathbf{x}_k - \eta_k \nabla_i f(\mathbf{x}_k) \mathbf{e}_i).$$

Using the Taylor expansion for a coordinate-wise L -smooth function, we have:

$$f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k) - \eta_k \frac{\partial f(\mathbf{x}_k)}{\partial x_i} \nabla_i f(\mathbf{x}_k) + \frac{L_i \eta_k^2}{2} (\nabla_i f(\mathbf{x}_k))^2.$$

4.2.2 Step 2: Taking Expectation

Taking expectation with respect to the random choice of i , we get:

$$\mathbb{E}[f(\mathbf{x}_{k+1})] \leq \mathbb{E}[f(\mathbf{x}_k)] - \eta_k \mathbb{E} \left[\frac{\partial f(\mathbf{x}_k)}{\partial x_i} \nabla_i f(\mathbf{x}_k) \right] + \frac{\eta_k^2}{2} \mathbb{E}[L_i (\nabla_i f(\mathbf{x}_k))^2].$$

Since i is chosen uniformly at random, we have:

$$\mathbb{E} \left[\frac{\partial f(\mathbf{x}_k)}{\partial x_i} \nabla_i f(\mathbf{x}_k) \right] = \frac{1}{n} \|\nabla f(\mathbf{x}_k)\|^2,$$

and

$$\mathbb{E}[L_i (\nabla_i f(\mathbf{x}_k))^2] = \frac{1}{n} \sum_{i=1}^n L_i (\nabla_i f(\mathbf{x}_k))^2.$$

4.2.3 Step 3: Applying the PL Condition

Using the PL condition, we have:

$$\frac{1}{2} \|\nabla f(\mathbf{x}_k)\|^2 \geq \mu(f(\mathbf{x}_k) - f^*).$$

Substituting this into our expectation inequality:

$$\mathbb{E}[f(\mathbf{x}_{k+1})] \leq \mathbb{E}[f(\mathbf{x}_k)] - \frac{\eta_k}{n} \|\nabla f(\mathbf{x}_k)\|^2 + \frac{\eta_k^2}{2n} \sum_{i=1}^n L_i (\nabla_i f(\mathbf{x}_k))^2.$$

4.3 Rate Derivation (With Constants)

Assume a constant step size $\eta_k = \eta \leq \frac{1}{L}$, where $L = \max_i L_i$. Then:

$$\mathbb{E}[f(\mathbf{x}_{k+1})] \leq \mathbb{E}[f(\mathbf{x}_k)] - \frac{\eta}{n} \|\nabla f(\mathbf{x}_k)\|^2 + \frac{\eta^2 L}{2n} \|\nabla f(\mathbf{x}_k)\|^2.$$

Simplifying, we get:

$$\mathbb{E}[f(\mathbf{x}_{k+1})] \leq \mathbb{E}[f(\mathbf{x}_k)] - \left(\frac{\eta}{n} - \frac{\eta^2 L}{2n} \right) \|\nabla f(\mathbf{x}_k)\|^2.$$

Using the PL condition:

$$\mathbb{E}[f(\mathbf{x}_{k+1})] - f^* \leq \left(1 - \frac{2\eta\mu}{n} + \frac{\eta^2 L\mu}{n} \right) (\mathbb{E}[f(\mathbf{x}_k)] - f^*).$$

For convergence, we need:

$$1 - \frac{2\eta\mu}{n} + \frac{\eta^2 L\mu}{n} < 1.$$

Solving for η , we get:

$$\eta < \frac{2n}{L}.$$

Thus, the algorithm achieves a convergence rate of:

$$\mathbb{E}[f(\mathbf{x}_T)] - f^* \leq \frac{C}{T},$$

where C is a constant depending on the initial conditions and parameters.

4.4 Special Cases

4.4.1 Case 1: Strongly Convex Functions

If f is strongly convex, the PL condition is satisfied with μ being the strong convexity constant. The convergence rate improves to linear convergence.

4.4.2 Case 2: High-Dimensional Functions

For high-dimensional functions, the coordinate-wise approach reduces the computational burden per iteration, making it more efficient than full gradient descent.